

Experiment Report: Checkout Button Colour Test

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Recommendation: DO NOT SHIP

Confidence: HIGH | Risk: LOW

- P(treatment best) = 8.3% - treatment likely worse
- Shipping would likely hurt revenue
- Expected value of shipping: \$-472.50/day vs keeping control: \$0.00/day

Experiment design

Field	Value
Hypothesis	Changing the CTA button from grey to green increases conversion rate
Variants	Control (Grey Button) vs Treatment (Green Button)
Sample size	10,000 control / 10,000 treatment
Duration	14 days
Significance level	alpha = 0.05

Frequentist results

Metric	Control	Treatment	Lift
Conversion rate	5.01%	4.59%	-8.38% rel.
95% CI	[4.60%, 5.46%]	[4.20%, 5.02%]	
Revenue / user	\$2.532	\$2.311	-8.73% rel.

Two-proportion z-test: $z = -1.389$, $p = 0.1647$ (not significant at $\alpha = 0.05$). Welch t-test (revenue/user): $t = -1.419$, $p = 0.1558$, Cohen's $d = -0.0201$. Power: 35.5% for a 10% MDE (required $n = 31,152/\text{group}$).

Bayesian results

Metric	Value
P(treatment > control)	8.3%
Expected relative uplift	-8.37%
Expected loss if shipping treatment	0.4315 pp
Expected loss if keeping control	0.0116 pp
Control 95% credible interval	[4.60%, 5.46%]
Treatment 95% credible interval	[4.20%, 5.02%]

Prior: Beta(1, 1)

Business impact

Assumptions: 5,000 daily visitors, 50% of traffic affected, \$45.00 revenue per conversion.

Metric	Value
Daily conversion lift	-10.5 conversions/day

Daily revenue lift	\$-472.50/day
Annual revenue lift (point estimate)	\$-172,462
Annual range (treatment CI)	\$-333,843 to \$3,239
EV of shipping vs status quo	\$-472.50/day

Scenario analysis

Scenario	Assumed lift	Daily revenue impact	Annual revenue impact	Probability	Recommendatio
Null (no effect)	+0.000%	\$0.00	\$0	5%	Do not ship
Pessimistic	-0.813%	\$-914.64	\$-333,843	10%	Do not ship
Base case	-0.420%	\$-472.50	\$-172,462	65%	Do not ship
Optimistic	+0.008%	\$8.87	\$3,239	20%	Continue testing

Caveats

- Annualised figures extrapolate a short test window; seasonality and novelty effects may erode the lift.
- Revenue impact assumes the conversion lift is the only effect (no change in average order value).
- Scenario probabilities are illustrative weights, not posterior probabilities.